

**Dualism**

The dualist theory argues that the mind (which is nonphysical) exists outside of the brain (which is physical) but that the two interact. The brain activity that happens as a result of the stimulus is associated with conscious perception, but the mind itself is separate.

**COULD ARTIFICIAL INTELLIGENCE BECOME CONSCIOUS?**

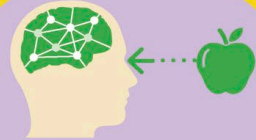
Some scientists do believe that artificial intelligence could be programmed to be conscious; others believe consciousness is not something that machines could ever learn.

The requirements of consciousness

The neural basis of consciousness remains an area of research, which is directed at identifying the structures and processes in the brain that are necessary to produce a conscious experience. The process of consciousness is believed to be at the level of individual neurons rather than at the level of individual molecules or atoms. It is likely that for consciousness to arise, the four factors below must be present.

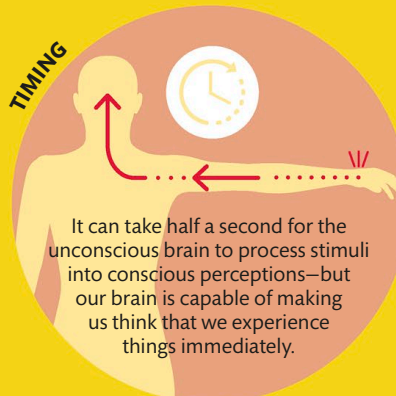
HIGH FIRING RATES**BETA BRAIN WAVES**

A normal state of consciousness occurs when neurons fire at fairly high rates. Beta waves (see p.42) occur when neurons fire at a high rate, and indicate alertness and logical, analytical thinking.

SYNCHRONOUS FIRING

Consciousness may depend on the synchronicity of neurons. Clusters of neurons firing in unison "bind" individual perceptions—such as sight, sound, and smell—to create one perception.

IN 1 OR 2 OF EVERY 1,000 MEDICAL PROCEDURES INVOLVING GENERAL ANESTHESIA, A PATIENT MAY BECOME CONSCIOUS

**TIMING**

It can take half a second for the unconscious brain to process stimuli into conscious perceptions—but our brain is capable of making us think that we experience things immediately.

FRONTAL ACTIVITY

The frontal lobes may play an important role in aspects of consciousness, including feelings of reflection, as well as coordinating levels of consciousness.